Permutations and Combinations

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Permutations Vs. Combinations

=> Both are counting techniques

=> We can solve many counting problems by

- => Finding the number of ways to arrange a specified number of distinct elements of a set of a particular size, where order of things matter.
- => Finding the number of ways to select a particular number of objects from a set of a particular size where order of things does not matter.

=> Example: - In how many ways can we select 3 students from a group of 5 students to stand in a line for a picture?

=> How many different committees of 3 students can be formed from a group of 4 students?

Permutations - definition

Definition

Let S be a set. A permutation is an **ordered** arrangements of the elements of some set S. - **ORDER** matters..

Example

Let $S = \{1, 2, 3\}$. The permutation of S is the different ordering of elements of set S. In the case they are:

- $(1, 2, 3), (1, 3, 2), (2, 1, 3), (2, 3, 1), (3, 1, 2), (3, 2, 1)$
- Note that: $(1, 2, 3) \neq (2, 1, 3)$ since order matter.
- An $r - permutation$ of a set is the **ordered** arrangement of r elements of the set. E.g. $(1, 2)$ is a $2 - permutation$ of the set S above.

The **number** of $r-$ *permutations* of a set

=> Note that we are now looking at the number of $r-$ *permutations* - **Counting**

=> Follow this discussion closely and note what this number means.

=> The number of $r-$ *permunations* is denoted by:

$$p(n, r)$$

=> Where n is the number of elements in the set.

=> Example: $p(9, 3)$ denotes the number of $3-$ *permutations* of a set with 9 elements

Finding $p(n, r)$ using the product rule

Example

=> Let $S = \{a, b, c\}$.

=> Find the number of $2 -$ *permutations* of set S.

Solutions

=> Counting manually -
$p(3, 2) = (a, b), (b, a), (a, c), (c, a), (b, c), (c, b) = 6$

=> Using the product rule: $p(3, 2) = 3 \times (3 - 1) = 3 \times 2 = 6$

- 3 ways to choose the first element, 2 ways to choose the second element

=> by product rule: $3 \times 2 = 6$

=> $p(55, 4) = 55 \times 54 \times 53 \times 52$ using the product rule.

Computing $p(n, r)$

Theorem

if n is a positive integer, and r is an integer such that, $1 \leq r \leq n$, then

$$p(n, r) = n \times (n-1) \times (n-2) \times \ldots \times (n-(r-1))$$

$$= n \times (n-1) \times (n-2) \times \ldots \times (n-r+1) = \frac{n!}{(n-r)!}$$

Case of r between 0 and n

If n is a positive integer and r is an integer with $0 \leq r \leq n$, then:

$$p(n, r) = \frac{n\,!}{(n-r)\,!}$$

$=>$ This is so since $p(n, 0) = \frac{n\,!}{n-r}\,! = \frac{n\,!}{n\,!} = 1$

Example

Example

Suppose *CSI*132 has 120 students. How many ways are there for 5 students in this class to give a presentations?

Solution

Solution

$\Rightarrow$ Task 1: is to select 1st student

$\Rightarrow$ Task 2: is to select 2nd student

$\Rightarrow$ Task 3: is to select 3rd student

$\Rightarrow$ Task 4: is to select 4th student

$\Rightarrow$ Task 5: is to select 5th student

- Using product rule: $120 * 119 * 118 * 117 * 116$ OR
- $p(120, 5) = \frac{120!}{(120-5)!} = \frac{120*119*118*117*116*115!}{115!}$

Example

Example

Suppose *CSI*132 has 120 students. How many ways are there for ALL students in this class to give a presentations?

Solution

Solution

$\Rightarrow$ This is giving by $p(120, 120)$

$\Rightarrow$ $\frac{120!}{(120-120)!} = \frac{120!}{0!} = \frac{120!}{1} = 120!$

Example

Example

Suppose that 30 students enter Miss UB.How many ways are there to choose the Queen, the 1st princess and the 2nd princess?

Exercise

Example

Supposes a salesperson has to visist 8 towns. She must begin her visit in a specified town, but can visit the other 7 towns in any order she wishes. How many possible orders can the salesperson use when visiting these towns?

solution - take home..

Exercise

Example

How many permutations of $\{a, b, c, d, e, f, g\}$ end with a?

solution - take home..

Combinations - Definition

- Counts possibilities if order does not matter. For example: $(a, b) = (b, a)$ and are counted as 1.
- $r-$combinations of a set is a subset of the set with r elements
- $\binom{n}{r}$ is the number of $r-$combinations of a set with n elements
- $\binom{n}{r} = \frac{n!}{r!(n-r)!}$ where n is a positive integer and $0 \leq r \leq n$
- Example: Let $S = \{a, b, c, d\}$
  - The $2-$combinations from S are $\{a, b\}, \{a, c\}, \{a, d\}, \{b, c\}, \{b, d\}, \{c, d\}$
  - so $\binom{4}{2} = 6$

number of $r-$combinations

Theorem

$$\binom{n}{r} = \frac{n!}{r!(n-r)!} \tag{1}$$

- n is a positive integer, r is an integer such that $0 \leq r \leq n$

More on $\binom{n}{r}$

$$\binom{n}{r} = \frac{p(n,r)}{p(r,r)} \tag{2}$$

- $\binom{4}{2} = \frac{4!}{2!(4-2)!} = 6$
- $\binom{4}{2} = \frac{p(4,2)}{p(2,2)} = \frac{12}{2} = 6$

proving that $\binom{n}{r} = \frac{n!}{r!(n-r)!}$

$$\binom{n}{r} = \frac{p(n,r)}{p(r,r)} \tag{3}$$

$$=> \quad \frac{\frac{n!}{(n-r)!}}{\frac{r!}{(r-r)!}}$$

$$=> \quad \frac{\frac{n!}{(n-r)!}}{r!} \text{ since } (r-r)! = 0! = 1$$

$$=> \quad \frac{n!}{r!(n-r)!}$$

Example

Example

$=>$ How many bit strings of length 10 contain exactly 4 $1's$?

Solution

$\Rightarrow$ The position of the $1's$ does not matter.

$\Rightarrow$ $\binom{10}{4} = \frac{10!}{4!(10-6)!} = \frac{10!}{4!*6!} = \frac{10*9*8*7*6!}{4!*6!}$

Example

Example

$=>$ How many bit strings of length 10 contain at most 4 $1's$?

Solution

At most 4 - 4 or less. Positions of $1's$ not significant.

$\Rightarrow$ Number of 10 bit strings with 0(zero) $1's$: $\binom{10}{0}$

$\Rightarrow$ Number of 10 bit strings with 1(one) $1's$: $\binom{10}{1}$

$\Rightarrow$ Number of 10 bit strings with 2(two) $1's$: $\binom{10}{2}$

$\Rightarrow$ Number of 10 bit strings with 3(three) $1's$: $\binom{10}{3}$

$\Rightarrow$ Number of 10 bit strings with 4(four) $1's$: $\binom{10}{4}$

Total: $\binom{10}{0} + \binom{10}{1} + \binom{10}{2} + \binom{10}{3} + \binom{10}{4}$

Example

Example

$=>$ How many bit strings of length 10 contain at least 4 $1's$?

Solution

At most 4 - The number of $1's$ can be $4, 5, 6, 7, 8, 9, 10$. Positions of $1's$ not significant.

$\Rightarrow$ Total $= \binom{10}{4} + \binom{10}{5} + \binom{10}{6} + \binom{10}{7} + \binom{10}{8} + \binom{10}{9} + \binom{10}{10}$ OR

$\Rightarrow$ Subtract from 2^{10} the number of 10 bit strings with $0, 1, 2, 3$ occurrences of $1's$.

$\Rightarrow$ $2^{10} - (\binom{10}{0} + \binom{10}{1} + \binom{10}{2} + \binom{10}{3})$

Exercise

Example

$=>$ How many bit strings of length 10 contain an equal number of $1's$ and $0's$?

Solution

It means those with 5 $1's$ and 5 $0's$.

$\Rightarrow$ Total =The number of 10 bit strings with five $1's$.

$\Rightarrow$ $\binom{10}{5} = \frac{10!}{5!*4!}$

Corollary

Theorem

$$\binom{n}{r} = \binom{n}{n-r} \tag{4}$$

=> Where n and r are positive integers.

=> Example, $\binom{100}{5} = \binom{100}{95}$

Proof of the corollary

$$\Rightarrow \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

$$\Rightarrow \binom{n}{n-r} = \frac{n!}{(n-r)!(n-(n-r))!} = \frac{n!}{r!(n-r)!}$$

Exercise of corollary

$\Rightarrow \binom{10}{3} = \binom{10}{7}$ by the corollary

$\Rightarrow \binom{10}{3} = \frac{10!}{3!7!}$

$\Rightarrow \binom{10}{7} = \frac{10!}{7!3!}$

$\Rightarrow$ so $\binom{10}{3} = \binom{10}{7}$

Binomial Coefficients

$\Rightarrow$ Recall $\binom{n}{r} = \binom{n}{n-r}$. we also did the proof for this.

$\Rightarrow$ Also called the binomial coefficient.

$\Rightarrow$ It appears as coefficient in the expansion of a binomial expression such as $(a+b)^n$

$\Rightarrow$ It allows for fast expansion of binomial expressions such as $(a+b)^n$.

Polynomial expansion

Consider $(x+y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$

$\Rightarrow$ This could be done by multiplying out the three terms

$$(x+y)(x+y)(x+y)$$

$\Rightarrow$ Or using combinatorial reasoning

$\Rightarrow$ We expect terms of the form x^3, x^2y, xy^2, y^3

Polynomial expansion - Using combinatorial reasoning

Consider $(x+y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$

$\Rightarrow$ This could be done by multiplying out the three terms

$$(x+y)(x+y)(x+y) = x^3 + [x^2y + x^2y + x^2y] + [xy^2 + xy^2 + xy^2] + y^3$$

$\Rightarrow$ To get x^3, x must be choosen in each of the sums. This can be done in only one way. This is $\binom{3}{0} = 1$.

$\Rightarrow$ To get x^2y an x must be chosen from 2 the sums, and y from the other sum. This can be done in $\binom{3}{1}$ ways $= 3$.

$\Rightarrow$ To get xy^2, an x is chosen from one of the sums, and two $y's$ from two sums. this can be done in $\binom{3}{2}$ ways $= 3$.

$\Rightarrow$ To get y^3, a y must be selected from each of the sums. This is done in $\binom{3}{3} = 1$ way.

The Binomial Theorem

$$(x+y)^n = \sum_{j=0}^{n} \binom{n}{j} x^{n-j} y^j =$$

$$\binom{n}{0} x^n + \binom{n}{1} x^{n-1} y + \ldots + \binom{n}{n-1} x y^{n-1} + \binom{n}{n} y^n$$

Binomial theorem

For

$$(a+b)^n = \sum_{r=0}^{n} \binom{n}{r} a^{n-r} b^r$$

we use binomial coefficients and then simplify each term.

Questions covered

- Expand $(a+b)^5$
- Expand $(x-4y)^4$
- Find the 7th term in $(3x^2-2y)^{13}$
- Find the required coefficients in two expansions

Expansion 1: $(a+b)^5$

$$(a+b)^5 = \sum_{r=0}^{5} \binom{5}{r} a^{5-r} b^r$$

So,

$$(a+b)^5 = \binom{5}{0} a^5 + \binom{5}{1} a^4 b + \binom{5}{2} a^3 b^2 + \binom{5}{3} a^2 b^3 + \binom{5}{4} a b^4 + \binom{5}{5} b^5$$

$$\boxed{(a+b)^5 = a^5 + 5a^4 b + 10a^3 b^2 + 10a^2 b^3 + 5ab^4 + b^5}$$

Expansion 2: $(x - 4y)^4$

$$(x - 4y)^4 = \sum_{r=0}^{4} \binom{4}{r} x^{4-r}(-4y)^r$$

Expanding term by term,

$$(x - 4y)^4 = x^4 + 4x^3(-4y) + 6x^2(16y^2) + 4x(-64y^3) + 256y^4$$

$$\boxed{(x - 4y)^4 = x^4 - 16x^3y + 96x^2y^2 - 256xy^3 + 256y^4}$$

7th term in $(3x^2 - 2y)^{13}$

General term:

$$T_{r+1} = \binom{13}{r}(3x^2)^{13-r}(-2y)^r$$

For the 7th term, $r + 1 = 7$, so $r = 6$.

$$T_7 = \binom{13}{6}(3x^2)^7(-2y)^6$$

$$= 1716 \cdot 3^7 \cdot 2^6 \, x^{14} y^6 = 1716 \cdot 2187 \cdot 64 \, x^{14} y^6$$

$$\boxed{T_7 = 240\,240\,384 x^{14} y^6}$$

Coefficient of a^5b^7 in $(a-2b)^{12}$

General term:

$$T_{r+1} = \binom{12}{r} a^{12-r}(-2b)^r$$

To get a^5b^7, we need

$$12 - r = 5 \quad \Rightarrow \quad r = 7$$

Hence the required coefficient is

$$\binom{12}{7}(-2)^7 = 792(-128)$$

$$\boxed{-101376}$$

Coefficient of $p^{16}q^7$ in $(3p^2 - 2q)^{15}$

General term:

$$T_{r+1} = \binom{15}{r}(3p^2)^{15-r}(-2q)^r$$

To get q^7, take $r = 7$. Then

$$(3p^2)^{15-7} = (3p^2)^8 = 3^8 p^{16}$$

So the coefficient is

$$\binom{15}{7}3^8(-2)^7 = 6435 \cdot 6561 \cdot (-128)$$

$$\boxed{-5\,406\,750\,720}$$

Example

Example

Using the binomial theorem, find the expansion of $(x + y)^6$

Example

Example

Using the binomial theorem, find the coefficient of $x^{12}y^{13}$ in the expansion of $(x+y)^{25}$

Example

Example

Using the binomial theorem, find the coefficient of $x^{12}y^{13}$ in the expansion of $(2x - 3y)^{25}$

Questions

Questions?